Annotated Bibliography: Palmer Penguins

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Annotated Bibliography: Palmer Penguins Dataset and Related Research

PHB 243b - Project 1 Reference Materials

This annotated bibliography provides background literature for the Palmer Penguins dataset, including the original ecological research, the data science education context, and broader ecological studies of Pygoscelis penguins in Antarctica.

Primary Data Sources

1. Original Ecological Study

Gorman KB, Williams TD, Fraser WR (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus Pygoscelis). PLOS ONE 9(3): e90081. https://doi.org/10.1371/journal.pone.0090081

This is the original research paper from which the Palmer Penguins dataset derives. The study examined sexual size dimorphism and sex-specific foraging niche partitioning among three penguin species (Adelie, Chinstrap, Gentoo) at Palmer Station, Antarctica, during 2007-2009. Methods included morphometric measurements (culmen length/depth, flipper length, body mass), molecular sexing using P2/P8 primers, and stable isotope analysis (carbon-13 and nitrogen-15) from blood samples to infer foraging ecology. Key findings: Chinstrap penguins showed the greatest sexual size dimorphism, followed by Gentoo and Adelie. Male Chinstrap and Gentoo penguins were enriched in nitrogen-15 relative to females, suggesting consumption of higher trophic-level prey. The study used information-theoretic model selection (AICc) for statistical inference. **Relevance:** Essential reading for understanding the biological context and data collection methods underlying the dataset.

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2. The palmerpenguins R Package Paper

Horst AM, Hill AP, Gorman KB (2022). Palmer Archipelago Penguins Data in the palmerpenguins R Package: An alternative to Anderson's Irises. The R Journal 14(1): 244-254. https://doi.org/10.32614/RJ-2022-020

The definitive paper introducing the palmerpenguins dataset for data science education. The authors make the case for replacing the iris dataset with Palmer penguins data, citing several advantages: (1) complete metadata and documentation versus iris's minimal provenance; (2) realistic missing values and unequal sample sizes that better prepare students for real-world data; (3) intuitive variable names (flipper length, bill depth) versus confusing botanical terms; (4) ethical concerns about iris's publication history in eugenics research. The paper demonstrates that penguins serves as a "near drop-in replacement" for iris across common educational use cases: data wrangling, visualization, linear modeling, PCA, and clustering. Downloaded over 462,000 times since July 2020 CRAN release. **Relevance:** Explains why this dataset was chosen for teaching and documents appropriate analytical approaches.

Broader Ecological Context

3. Climate Change and Penguin Populations

Clucas GV, Dunn MJ, Dyke G, et al. (2014). A reversal of fortunes: climate change 'winners' and 'losers' in Antarctic Peninsula penguins. Scientific Reports 4: 5024. https://doi.org/10.1038/srep05024

A landmark study on how climate change differentially affects the three Pygoscelis species. Using molecular techniques to assess demographic history since the Last Glacial Maximum (LGM), the researchers found that all three species initially benefited from post-LGM warming by expanding from glacial refugia. However, current anthropogenic warming has caused a "reversal of fortunes": Gentoo penguins continue to expand as climate "winners," while Adelie and Chinstrap penguins have become "losers" with declining populations. The mechanism involves decreased sea ice, loss of winter habitat, and reduced krill availability. This decoupling of historic responses from current trends suggests anthropogenic impacts outside the range of natural variation. **Relevance:** Provides ecological context for species differences observed in the dataset and motivation for conservation research.

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4. Global Chinstrap Population Assessment

Strycker N, Wethington M, Borowicz A, et al. (2020). A global population assessment of the Chinstrap penguin (Pygoscelis antarctica). Scientific Reports 10: 19474. https://doi.org/10.1038/s41598-020-76479-3

The first comprehensive global census of Chinstrap penguins. Using satellite imagery, drone imagery, and ground counts, researchers estimated 3.42 million breeding pairs across 375 extant colonies. Twenty-three previously known colonies were found absent or extirpated, while 26 new or unreported colonies were identified. Among colonies with historical data from the 1980s, 45% have declined and only 18% have increased. **Relevance:** Demonstrates modern methods for large-scale ecological assessment and provides population context for the Chinstrap penguins in the Palmer dataset.

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5. Gentoo Population Genetics

Peña MF, Poulin E, Dantas GPM, González-Acuña D, Petry MV, Vianna JA (2014). Have historical climate changes affected Gentoo penguin (Pygoscelis papua) populations in Antarctica? PLOS ONE 9(4): e95375. https://doi.org/10.1371/journal.pone.0095375

This study examined how climate change over the past 50 years in the West Antarctic Peninsula (WAP) has affected Gentoo penguin populations. The atmospheric temperature increase and changes in sea-ice dynamics have led to population expansion of sub-Antarctic Gentoo penguins and retreat of Adelie penguins. Using mitochondrial DNA analysis (HVRI and β-fibrinogen intron 7), researchers traced demographic history and found signatures of population expansion beginning approximately 13,000 BP, suggesting recolonization from peri-Antarctic refugia after ice sheet retreat. **Relevance:** Provides genetic context for species differences observed in the Palmer dataset and demonstrates molecular ecology methods relevant to biostatistics training.

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6. Trophic Interactions and the Sea Ice Hypothesis

Cimino MA, Lynch HJ, Saba VS, Oliver MJ (2016). Projected asymmetric response of Adelie penguins to Antarctic climate change. Scientific Reports 6: 28785. https://doi.org/10.1038/srep28785

Models project that Adelie penguin populations will respond asymmetrically to continued climate change, with colonies in different regions showing divergent trajectories depending on local sea ice conditions. The study used species distribution models to forecast population changes under climate scenarios. **Relevance:** Demonstrates predictive modeling approaches that could be applied to morphometric relationships in the Palmer dataset.

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7. Adélie Penguin Morphometric Sexing

Polito MJ, Trivelpiece WZ (2008). Transition to independence and evidence of extended parental care in the gentoo penguin (Pygoscelis papua). Marine Biology 154: 231-240.

Ainley DG, Emison WB (1972). Sexual size dimorphism in Adélie penguins. Ibis 114: 267-271.

Classic papers establishing morphometric approaches to penguin sexing. These studies demonstrated that bill dimensions and body mass can reliably discriminate sex in Pygoscelis penguins, laying the groundwork for the molecular validation in Gorman et al. (2014). Logistic regression models using culmen measurements achieve high classification accuracy. **Relevance:** Provides methodological background for using morphometric predictors and validates the discriminant analysis applications common in educational uses of the Palmer dataset.

Data Science and Statistics Education

8. Release Announcement

Hill AP (2020). Release the penguins. RStudio Education Blog, July 28, 2020. https://education.rstudio.com/blog/2020/07/palmerpengu
cran/

The official announcement of the palmerpenguins package on CRAN. Describes the motivation for creating the package, highlights key features (missing values, Simpson's Paradox examples, multiple factor variables), and provides example visualizations. Notes that the dataset is "a great intro dataset for data exploration & visualization" suitable as an iris alternative. **Relevance:** Documents the educational intent and provides introductory examples.

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9. Simpson's Paradox Example

Pearl J (2014). Comment: Understanding Simpson's Paradox. The American Statistician 68(1): 8-13. https://doi.org/10.1080/00031305.2014.876829

Pearl's paper provides a causal inference framework for understanding Simpson's Paradox, the phenomenon whereby an association between two variables reverses sign when conditioned on a third variable. While Pearl's paper addresses the paradox theoretically (not the penguin data specifically), the Palmer Penguins dataset offers a pedagogically useful illustration: the overall correlation between bill depth and body mass is negative, but within each species the correlation is positive. This reversal occurs because Gentoo penguins have larger bodies but shallower bills than Adelie penguins, and species acts as a confounding variable. **Relevance:** Pearl's causal framework combined with this accessible dataset provides an effective teaching example for confounding and the importance of stratified analysis.

Machine Learning Applications

10. Species Classification Studies

Various authors (2020-2025). Penguin species classification using machine learning. Multiple sources including Kaggle notebooks and GitHub repositories (grey literature).

- https://www.kaggle.com/code/freddymoreno/using-machine-learning-to-classify-penguin-species
- https://github.com/ketanmakde/DT-RF_Penguin-Data-Antarctica
- https://github.com/hussaifm/penguins-dataset-R

Educational implementations of classification algorithms. Multiple tutorials demonstrate species classification using the Palmer dataset with decision trees, random forests, logistic regression, k-nearest neighbors, and neural networks. The Gentoo species is easily separable based on culmen depth; Adelie and Chinstrap show more overlap but can be distinguished by culmen length. Note: Some tutorials report very high accuracy (e.g., 100% for random forests), which likely reflects overfitting to small datasets rather than generalizable performance; this provides a useful teaching opportunity regarding cross-validation and model evaluation. **Relevance:** Demonstrates that the dataset supports machine learning education beyond traditional statistical methods, while also illustrating common pitfalls in model assessment.

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11. Body Mass Prediction (Regression)

Scikit-learn MOOC (2021-2025). The penguins datasets. INRIA. https://inria.github.io/scikit-learn-mooc/python_scripts/trees_datas

The scikit-learn massive open online course uses Palmer penguins to teach regression concepts, specifically predicting body mass from flipper length. Random forest models achieve RMSE of approximately 317g and R-squared of 0.877. The dataset demonstrates clear positive relationships between morphometric variables that serve as intuitive examples for linear and tree-based regression. **Relevance:** Validates the regression analyses planned for Project 1.

Citation for the Dataset

When using the Palmer Penguins data, cite both the R package and the original data collection:

Data package: Horst AM, Hill AP, Gorman KB (2020). palmerpenguins: Palmer Archipelago (Antarctica) penguin data. R package version 0.1.0. https://allisonhorst.github.io/palmerpenguins/ DOI: 10.5281/zenodo.3960218

Original data collection: Gorman KB, Williams TD, Fraser WR (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus Pygoscelis). PLOS ONE 9(3): e90081. https://doi.org/10.1371/journal.pone.0090081

Data repository: Data originally published through the Environmental Data Initiative (EDI) Data Portal, available under CC0 license in accordance with the Palmer Station LTER Data Policy.

Video Tutorials and Screencasts

12. TidyTuesday Week 31 (2020): Palmer Penguins

R for Data Science Online Learning Community (2020). TidyTuesday 2020 Week 31: Palmer Penguins. GitHub repository, July 28, 2020. https://github.com/rfordatascience/tidytuesday/tree/master/data/2020/2020-07-28

The Palmer Penguins dataset was featured as the TidyTuesday challenge for Week 31 of 2020. TidyTuesday is a weekly social data project organized by the R for Data Science Online Learning Community, where participants analyze a common dataset and share visualizations. This week generated numerous high-quality submissions demonstrating diverse approaches to exploratory data analysis and visualization with the penguins data. **Relevance:** Provides access to community-contributed code examples and visualizations that can serve as inspiration for student projects.

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13. Tidymodels Classification Tutorial (Video)

Silge J (2020). Get started with tidymodels and #TidyTuesday Palmer penguins. Blog post and YouTube screencast, July 28, 2020. https://juliasilge.com/blog/palmer-penguins/ YouTube: https://www.youtube.com/watch?v=z57i2GVcdww

Julia Silge, software engineer at Posit (formerly RStudio) and tidymodels maintainer, demonstrates building classification models to predict penguin sex from morphometric measurements. The tutorial covers the complete tidymodels workflow: data splitting with stratification, recipe specification, model fitting (logistic regression and random forest), bootstrap resampling for model evaluation, and interpretation of results. A key pedagogical insight: the simpler logistic regression model performed comparably to random forest, illustrating the principle of parsimony. **Relevance:** Excellent introduction to the tidymodels framework using familiar data; demonstrates that simpler models often suffice when interpretability matters.

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14. Advanced ggplot2 Visualization

Scherer C (2020). TidyTuesday 2020/31: Palmer Penguins visualization. GitHub and Twitter, July 28, 2020. https://github.com/z3tt/TidyTuesday/blob/main/R/2020_31_PalmerPenguins.Rmd https://twitter.com/CedScherer

Cedric Scherer, data visualization specialist, created publication-quality visualizations for the Palmer Penguins TidyTuesday challenge, including innovative boxplot variants and raincloud plots. His code demonstrates advanced ggplot2 techniques: custom themes, color palettes, annotations, and combining multiple plot types. Scherer also developed a hands-on tutorial for OutlierConf 2021 (“ggplot Wizardry”) that uses the penguins dataset to teach advanced visualization techniques. **Relevance:** Demonstrates professional-quality data visualization techniques; code is fully reproducible and well-documented.

R-Bloggers Tutorials

15. Basic Data Analysis with palmerpenguins

R-bloggers (2020). Basic data analysis with palmerpenguins. July 10, 2020. https://www.r-bloggers.com/2020/07/basic-data-analysis-with-palmerpenguins/

An introductory tutorial covering exploratory data analysis routines with the penguins dataset, including correlation analysis, parallel coordinate plots, and basic classification using support vector machines. The post emphasizes that the penguin dataset allows practice with pre-processing workflows typical of real-world data. **Relevance:** Provides a straightforward EDA template suitable for students beginning their analysis.

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16. Penguins Dataset Overview

R-bloggers (2020). Penguins Dataset Overview: iris alternative in R. June 17, 2020. https://www.r-bloggers.com/2020/06/penguins-dataset-overview-iris-alternative-in-r/

Early coverage of the palmerpenguins package, discussing the motivation for moving away from the iris dataset due to Ronald Fisher's eugenicist past. The post provides initial visualizations and notes that Allison Horst packaged the data under CC-0 license. **Relevance:** Documents the ethical considerations that motivated adoption of this dataset in the R community.

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17. Hyperparameter Optimization Tutorial

R-bloggers (2022). Hyperparameter Optimization on the Palmer Penguins Data Set. November 10, 2022. https://www.r-bloggers.com/2022/11/hyperparameter-optimization-on-the-palmer-penguins-data-set/

Demonstrates hyperparameter tuning for classification trees using the mlr3 framework. The tutorial introduces tuning spaces and automated machine learning concepts, optimizing tree depth, minimum split size, and other parameters. Uses cross-validation to assess generalization performance. **Relevance:** Extends beyond basic modeling to demonstrate principled approaches to model selection and tuning.

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18. Classification Comparison: Logistic Regression vs. Random Forest

R-bloggers (2022). Classification: Logistic Regression and Random Forest. February 2022. https://www.r-bloggers.com/2022/02/classification-logistic-regression-and-random-forest/

A comparative tutorial predicting penguin sex using logistic regression and random forest algorithms within the tidymodels framework. Key finding: logistic regression can outperform more complex tree-based methods on this dataset, reinforcing the value of starting with simpler models. The tutorial covers stratified splitting, model specification, and performance comparison. **Relevance:** Demonstrates that model complexity does not guarantee better performance; supports the course emphasis on interpretable models.

Artwork and Teaching Resources

19. palmerpenguins Artwork

Horst A (2020). Art for teaching with palmerpenguins. https://allisonhorst.github.io/palmerpenguins/articles/art.html

Allison Horst created original artwork to accompany the palmerpenguins package, including illustrations explaining bill measurement methods (culmen length and depth), species identification guides, and the package hex logo featuring all three penguin species. The artwork is released under permissive licensing for educational use with attribution. **Relevance:** Visual resources that enhance teaching effectiveness; particularly useful for explaining measurement methods and species differences.

Summary Table

Reference	Type	Key Contribution
Gorman et al. 2014	Original research	Source of morphometric data
Horst et al. 2022	R Journal article	Dataset documentation, iris comparison
Clucas et al. 2014	Ecology	Climate winners/losers framework
Strycker et al. 2020	Population ecology	Global Chinstrap census methods
Hill 2020	Education	Package release, teaching applications
Silge 2020	Video tutorial	Tidymodels classification workflow
Scherer 2020	Visualization	Advanced ggplot2 techniques
R-bloggers 2020-2022	Tutorials	EDA, classification, hyperparameter tuning
Horst 2020	Artwork	Teaching illustrations

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